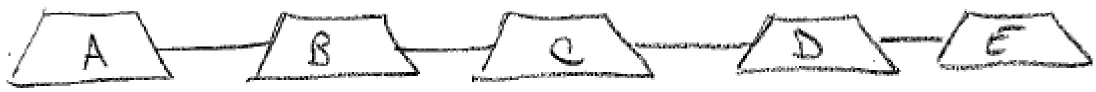
ECE4273 **EXTRA CREDIT HOMEWORK** **Due: 12/8/2025**

NOTE: No late homework will be accepted for this assignment!

1. (a) In an optical line terminal, give two possible functions that a transponder may be used for.
(b) Why would you want to use a line terminal at the boundary between networks operated by two different companies?

2. Consider the five-node linear network shown below. You are required to set up the following lightpaths: A-B, B-C, C-D, C-E, A-C, D–E, and B-D. You have three wavelengths to use, l_1, l_2, and l_3. You have available three types of OADMs – one which add/drops l_1 and l_2, one which add/drops l_1 and l_3, and one which add/drops l_2 and l_3. Determine which OADMs are needed at each node to construct the network.



3. (a) What are the benefits and costs of using an optical core for an optical cross connect instead of an electronic core? (b) Why might you choose to use a transponder with an optical core?

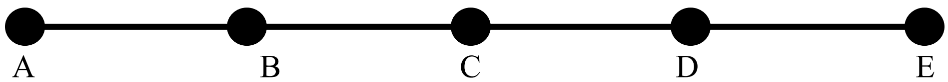
4. (a) One of the traffic models we looked at was the statistical model. Why would knowing the detailed statistics of the traffic help in designing the network connections? (b) Under what conditions might you use the fixed traffic model for a network design?

5. In the maximum load model, we try to limit the maximum load on any one connection by spreading the traffic across the network. (a) Why might this be beneficial to the operation and/or design of the network? (b) What are some practical limitations to the application of the maximum load model to a real network design? (c) Describe two issues that make routing and wavelength assignment such a difficult problem.

6. (a) What are the fundamental differences between a PWDM, hub, and all-optical network design? (b) For a given traffic pattern or application, is the all-optical design always favorable over the hub design? Why or why not?

7. You are asked to perform routing and wavelength assignment on a five-node line network. Assume that one fiber is used for traffic heading left to right, and one fiber is used for traffic heading right to left. The traffic matrix for the network is as follows (next page):

	A	B	C	D	E
A	--	1	1	0	1
B	1	--	2	1	0
C	1	2	--	1	0
D	0	1	1	--	1
E	1	0	0	1	--



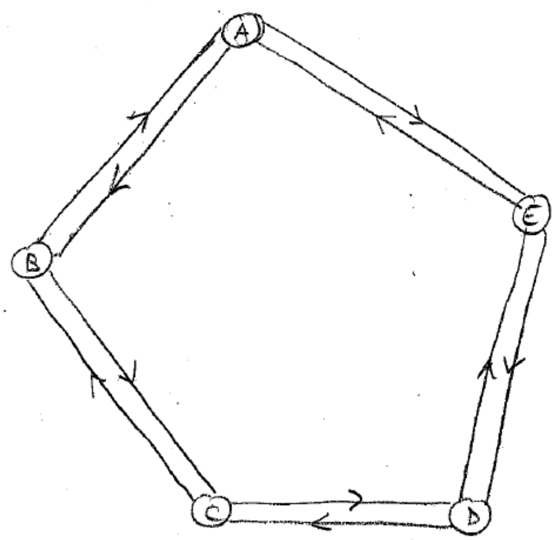
(a) Using the greedy algorithm, draw the set of connections for the network, assigning the lightpaths by going through the table starting at top left, working left to right and top to bottom, like we did in class. How many wavelengths do you need?

(b) Repeat part (a) by working through the table in a different order than in part (a). Does the number of wavelengths required by the network change?

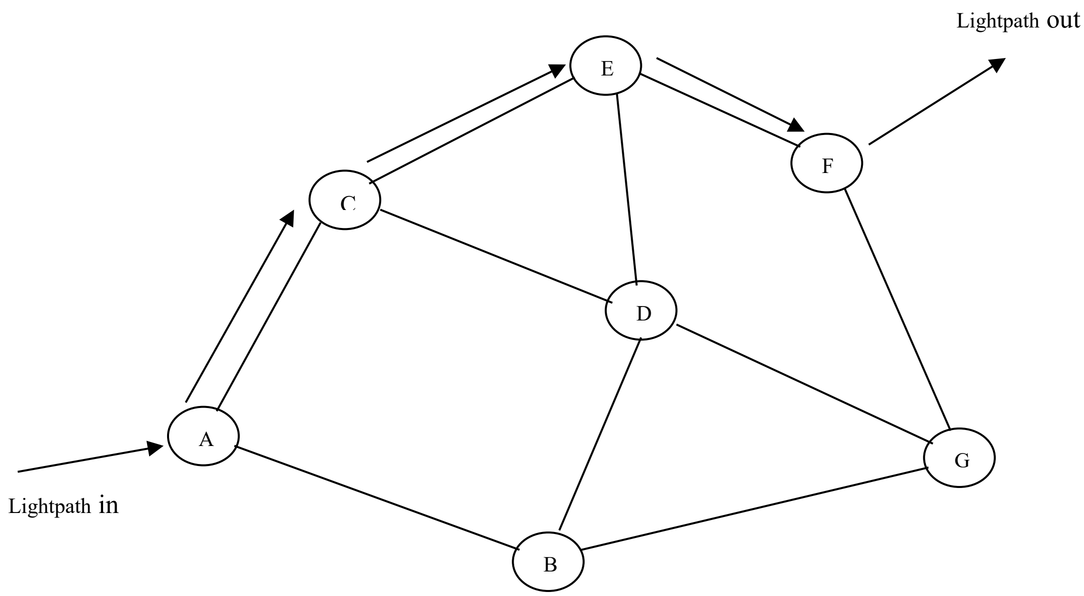
(c) Would the use of a ring network reduce the number of wavelengths used? For the ring network, simply add a connection between node A and node E. Explain your answer.

8. A ring network has the traffic distribution shown in the table below. Implement the network using a PWDM architecture and determine the number of wavelengths needed in the network, assuming the lightpaths are requested in the order they are listed.

A ←→ B 0.75
A ←→ C 1.50
C ←→ E 0.25
B ←→ E 0.50
D ←→ B 1.00
D ←→ A 0.25
C ←→ B 0.50



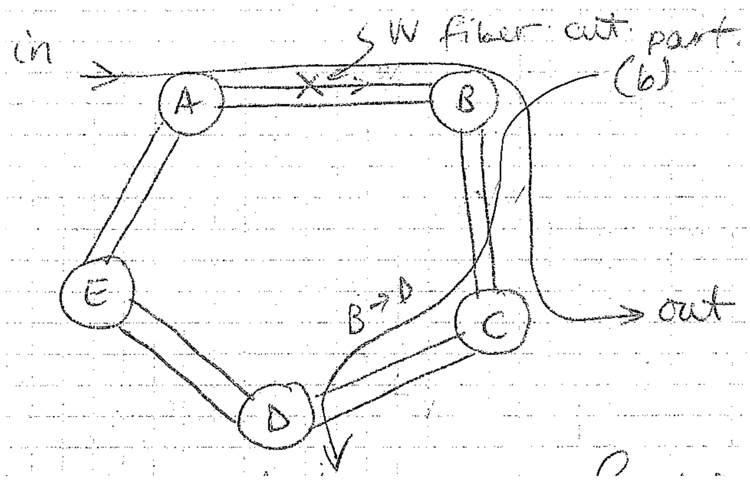
9. Consider the network shown, which supports a path from node A to node F, as shown in the picture. (a) How would the network re-route the signal if all the connections between nodes E and F were physically cut, and line protection was used? (b) How would the network re-route the signal if path protection is in use? (c) What are one advantage and one disadvantage of using line protection compared to path protection?



10. What would be the benefits and costs of using (a) BLSR/4 protection scheme over a BLSR/2 scheme? (b) a BLSR/2 protection scheme over a UPSR protection scheme?

11. A point-to-point link supporting four working lines uses 1:N protection. The priorities are set such that Working line 1 (W1) is of higher priority than Working line 2 (W2), denoted W1 > W2, and that W2 > W3 > W4. We assume that low-priority data is switched onto the protection line when the protection line is not otherwise in use. Assume automatic reversion is in use. Describe in detail what would happen when (a) W3 fails between the nodes. (b) W3 is fixed, then a little later W1 fails. (c) W3 has failed, then W4 fails before W3 is fixed. (d) W3 has failed, and W1 fails before W3 is fixed. (e) What might happen differently if automatic reversion is not used?

12. The ring network below (next page) has two fibers connecting each node. A lightpath is routed from node A to node C. A failure on the connection between nodes A and B occurs that cuts the working fiber. (a) Compare the protection path that is taken for a UPSR and a BLSR/2 protection scheme implemented on the ring. How are they different or the same? (b) A second lightpath exists on the ring connecting nodes B and D. Which of the methods (UPSR or BLSR/2) could correct for a failure on this lightpath?



13. (a) Describe one advantage and one disadvantage of using distributed management over centralized management in an optical network. (b) What is the purpose of an ONU and an NNU in a management network? (c) Give two factors that might drive the decisions on how many units of a specific piece of equipment (say an OXC) to purchase and where they should be located within (or in reserve for) the network.